23/01/2026
Morning

Aakash
Medical | IIT-JEE | Foundations

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Memory Based
Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2026 (Online) Phase-1

(Physics, Chemistry and Mathematics)





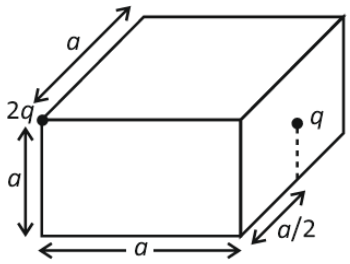
PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. There are two point charges, one at vertex and other at face as shown the cube. Find electric flux through the cube.



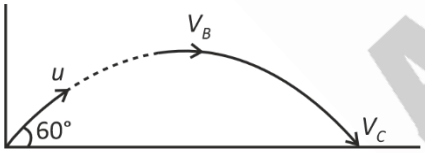
(1) $3q/\varepsilon_0$ (2) q/ε_0

(3) $3q/4\varepsilon_0$ (4) $5q/\varepsilon_0$

Answer (3)

Sol. $\phi = \frac{q_{enc}}{\varepsilon_0} = \frac{\frac{2q}{8}+\frac{q}{2}}{\varepsilon_0} = \frac{3q}{4\varepsilon_0}$

2. If a projectile is being projected with speed v at an angle 60° with horizontal. Find the ratio of speed at highest point (V_B) to the speed at final point (V_C).



(1) 3 : 4 (2) 1 : 3

(3) 1 : 2 (4) 1 : 12

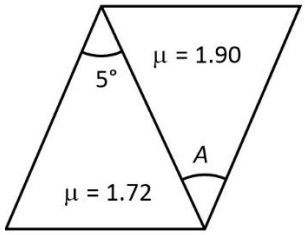
Answer (3)

Sol. $V_C = v$

$V_B = v\cos 60° = \frac{v}{2}$

$\frac{V_B}{V_C} = \frac{1}{2}$

3. Find A for dispersion without deviation.



(1) 3 (2) 4

(3) 4.5 (4) 5

Answer (2)

Sol. $A_1(\mu_1 - 1) = A_2(A_2 - 1)$

$5(0.72) = A(.9)$

$\Rightarrow A = 4$

4. For the given set of measurement find relative error.

20.00, 19.75, 18.25, 17.01

(1) 0.12 (2) 0.06

(3) 0.09 (4) 0.17

Answer (2)

Sol. $\bar{x} = \frac{20.00+18.25+19.75+17.01}{4}$

$\Rightarrow \bar{x} = 18.75$

$\Rightarrow |\Delta\bar{x}| = \frac{1.25+0.5+1+1.74}{4} = 1.12$

$\frac{\Delta\bar{x}}{\bar{x}} = \frac{1.12}{18.75} = 0.06$



5. Find out the correct energy for the ground state or energy transition. (symbols have usual meaning & $n \to m$ gives the transition)

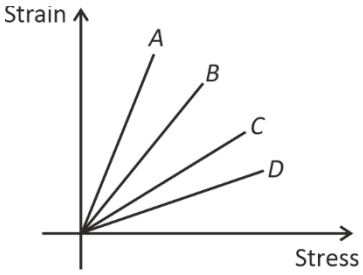
(1) H (–6.8 eV) (2) Li^{2+} (–13.6 eV)

(3) $He^{+}_{2\to1}$ (40.8 eV) (4) $Be^{3+}_{2\to1}$ (+13.6 eV)

Answer (3)

Sol. $E_n = 13.6\,\frac{z^2}{n^2}$

6. Which of the following material has bigger Young's modulus



(1) *A* (2) *B*

(3) *C* (4) *D*

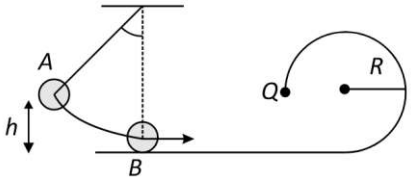
Answer (4)

Sol. $\text{Slope} = \frac{\text{Strain}}{\text{Stress}}$

$Y = \frac{1}{\text{Slope}}$

Thus *D*

7. Find *h* (in m) such that ball *B* just be able to reach *Q* after elastic collision with A. Mass of both the bodies are same m. (R = 10 cm)



(1) $\frac{3}{5}$ (2) $\frac{1}{\sqrt{5}}$

(3) $\frac{1}{4}$ (4) $\frac{1}{5}$

Answer (3)

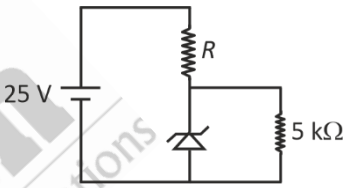
Sol. $V_A = V_B$, $V_A = \sqrt{5gR}$

$m_A gh = \frac{1}{2} m_A v_A^2$

$h = \frac{5}{2}R$

$h = \frac{5}{2}R = \frac{5}{2} \times \frac{1}{10} = \frac{1}{4}$

8. For the given circuit the breakdown voltage of Zener diode is V_z = 5 volts. And it can with stand maximum current of I_Z = 5 mA.



Find the value of *R*.

(1) $\frac{10}{3}$ kΩ (2) 5 kΩ

(3) $\frac{15}{4}$ kΩ (4) 8 kΩ

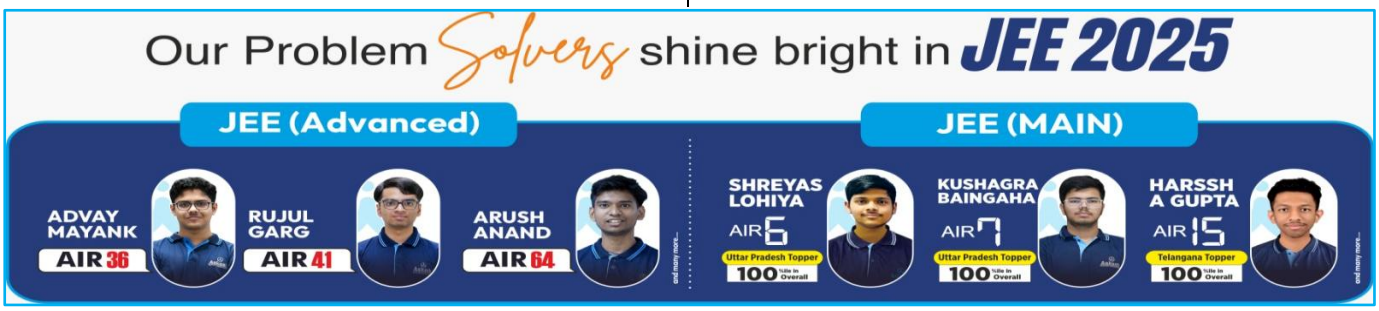
Answer (1)

Sol. I_1 across 6 kΩ is

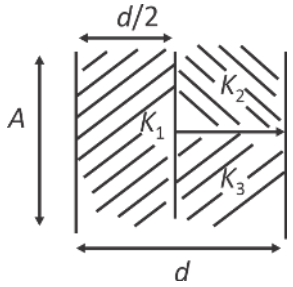
$I_1 = \frac{5}{5000} = 1$ mA

So, total I = 6 mA

$R = \frac{20}{6} \times 1000 = \frac{10}{3}$ kΩ



9. Find capacitance of capacitor given below if each plate has area A and separation is d between them.



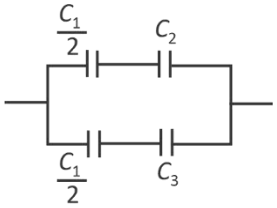
$K_1 = 3 \quad K_2 = 5 \quad K_3 = 2$

(1) $\frac{123\varepsilon_0 A}{40d}$ (2) $\frac{140\varepsilon_0 A}{6d}$

(3) $\frac{120\varepsilon_0 A}{30d}$ (4) $\frac{128\varepsilon_0 A}{4d}$

Answer (1)

Sol.



$C_1 = \frac{6\varepsilon_0 A}{d} \quad C_2 = \frac{5\varepsilon_0 A}{d} \quad C_3 = \frac{2\varepsilon_0 A}{d}$

$C_{eq} = \frac{123\varepsilon_0 A}{40d}$

10. A simple pendulum of length 3 m completes 20 oscillations in 10 sec. What is length of another pendulum which completes 40 oscillations in same times

(1) 0.75 m (2) 1.5 m

(3) 12 m (4) 6 m

Answer (1)

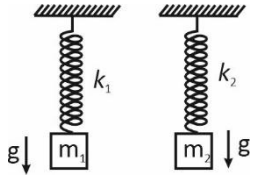
Sol. $T = 2\pi\sqrt{\frac{l}{g_{eff}}}$

$\frac{Y_2}{Y_4} = \frac{\sqrt{3}}{\sqrt{l}}$

$l = \frac{3}{4}\text{m}$

11. Find ratio of energy stored in the two springs as shown in figure below

Given: $\frac{K_1}{K_2} = \frac{1}{2}$ and $\frac{m_1}{m_2} = \frac{1}{4}$



(1) $\frac{1}{4}$ (2) $\frac{1}{2}$

(3) $\frac{1}{8}$ (4) 1

Answer (3)

Sol. Given : $\frac{k_1}{k_2} = \frac{1}{2}$ and $\frac{m_1}{m_2} = \frac{1}{4}$

$E = \frac{2kq}{r^2} \Rightarrow E_{net} = E\sqrt{3}$

12. In two different YDSE setup, two different monochromatic wave are used but fringe width on the screen is same. If $\frac{\lambda_1}{\lambda_2} = \frac{1}{2}$ and $\frac{d_1}{d_2} = \frac{2}{1}$, then find the ratio of $\frac{D_1}{D_2}$ (All symbols are standard).

(1) 1 (2) $\frac{1}{2}$

(3) 2 (4) 4

Answer (4)

Sol. $\Delta\omega = \frac{\lambda D}{d}$

So, $\frac{\lambda_1 D_1}{d_1} = \frac{\lambda_2 D_2}{d_2}$

$\frac{D_1}{D_2} = \left(\frac{d_1}{d_2}\right)\cdot\left(\frac{\lambda_2}{\lambda_1}\right) = \left(\frac{2}{1}\right)\times\left(\frac{2}{1}\right)$

$\frac{D_1}{D_2} = 4$



13. A 25 kg mass moving with 30 m/s and another mass 15 kg of same material moving with 10 m/s in opposite direction, collides perfectly inelastically. Find the rise in temperature of the system.

(Given: $C = 5 \times$ cal/g°C & 1 cal = 4.2 J)

(1) $\frac{1}{50}$ °C

(2) $\frac{1}{112}$ °C

(3) $\frac{1}{150}$ °C

(4) $\frac{1}{125}$ °C

Answer (1)

Sol. $25 \times 30 - 15 \times 10 = 40\ V_f \Rightarrow V_f = 15$ m/s

$$K\varepsilon_i - K\varepsilon_f = \frac{1}{2}25\times900+\frac{1}{2}\times15\times100-\frac{1}{2}40(225) = 7500 \text{ J}$$

$$ms\Delta T = 7500 \Rightarrow \Delta T = \frac{7500}{40\times5000\times4.2} = \frac{1}{112} \text{ °C}$$

14. Two point particles of masses $2m$ and m are attached to ends of a massless rod of length l. Find Angular momentum of this system about an axis passing through their center of mass and perpendicular to the rod if the system is rotating with angular velocity ω about the axis.

(1) $\frac{2}{3}m\omega l^2$

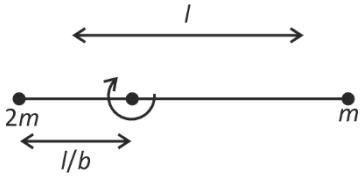
(2) $\frac{1}{3}m\omega l^2$

(3) $\frac{m\omega l^2}{9}$

(4) $m\omega l^2$

Answer (1)

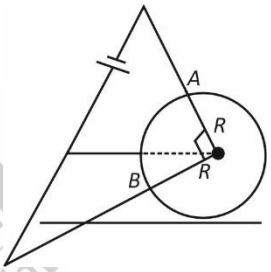
Sol.



$$L = 2m\omega\left(\frac{l}{3}\right)^2 + m\omega\left(\frac{20}{3}\right)^2$$

$$L = \frac{2}{9}m\omega l^2 + \frac{4}{9}m\omega l^2 = \frac{2}{3}m\omega l^2$$

15. A uniform wire is bent into the shape of a circle of radius R has resistance per unit length = λ. A battery is connected across point A and B which substends an angle of 90° at its centre as shown. Find the equivalent resistance of this circuit.



(1) $\frac{3\pi}{4}\lambda R$ (2) $\frac{\pi}{2}\lambda R$

(3) $\frac{3\pi}{8}\lambda R$ (4) $\frac{\pi}{4}\lambda R$

Answer (3)

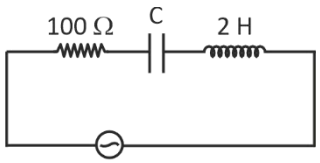
Sol. $R_1 = \lambda\frac{3\pi R}{2}; R_2 = \frac{\lambda\pi}{2}R$

$$R_{eq} = \frac{\lambda\frac{\pi}{2}R\,\lambda\frac{3\pi}{2}R}{\pi\frac{\pi}{2}R+\lambda\frac{3\pi}{2}R} = \frac{\lambda^2\frac{3\pi^2}{4}R^2}{\lambda\times2\pi R}$$

$$R_{eq} = \frac{3\pi}{8}\lambda R$$



16. The peak current in given LCR series Ac circuit shown is 5 mA then capacitance in μF is nearly.



$\varepsilon = 50\sin 100t$

(1) 1 (2) 3

(3) 4 (4) 5

Answer (1)

Sol. $5\times10^{-3} = \dfrac{50}{\sqrt{(100)^2 + (200 - X_C)^2}}$

$10^4 + (200 - X_C)^2 = 10^8$

$(200 - X_C) \approx \pm 10^4$

$X_C \approx 10^4 = \dfrac{1}{\omega C}$

$C = 10^{-6}$

17. Two media of refractive indices n_1 & n_2 have plane interface. Speed of light in 1st is 2.4×10^8 m/s & in 2nd is 2.8×10^8 m/s.

Find critical angle of incidence when light travels from 1st to 2nd.

(1) $\sin^{-1}\left(\dfrac{6}{7}\right)$ (2) $\sin^{-1}\left(\dfrac{7}{8}\right)$

(3) $\sin^{-1}\left(\dfrac{8}{9}\right)$ (4) $\tan^{-1}\left(\dfrac{9}{8}\right)$

Answer (1)

Sol. $\dfrac{n_1}{n_2} = \dfrac{2.8\times10^8}{2.4\times10^8} = \dfrac{1}{\sin(\theta_c)}$

$\Rightarrow \quad \theta_c = \sin^{-1}\left(\dfrac{6}{7}\right)$

18. **Assertion (A):** In a ferromagnetic material, magnetic dipole are aligned in a localized domain.

Reason (R): When ferromagnetic substance is heated above the curie temperature, its magnetic properties are weakened heavily.

(1) A is correct, R is correct and correct explanation of A

(2) A is correct, R is correct but not correct explanation of A

(3) Both A and R are incorrect

(4) A is correct, R is incorrect

Answer (2)

Sol. Conceptual.

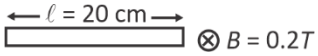
19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Find emf induced in Rod after it has fallen through 2000 *m* under gravity (in volt).



Answer (8)

Sol. $\varepsilon = Bvl = 0.2\times\sqrt{2gh}\times l$

$\Rightarrow$ 8 volt

22.

23.

24.

25.



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. The correct order of ionisation energy of

   Cl, S, P, Al, Si is

   (1) Cl > P > S > Si > Al

   (2) P > Cl > S > Al > Si

   (3) Cl > S > P > Si > Al

   (4) Cl > Al > Si > P > S

Answer (1)

Sol. **kJ/mol**

Cl ⇒ 1256

S ⇒ 999

P ⇒ 1012

Al ⇒ 577

Si ⇒ 786

2. Consider the statements below :

   Statement I : $[CoBr_4]^{2-}$ absorbs lesser energy than $[CoCl_4]^{2-}$.

   Statement II : $[CoCl_4]^{2-}$ has higher crystal field splitting energy than $[CoBr_4]^{2-}$.

   (1) Both statement I and statement II are correct

   (2) Both statement I and statement II are incorrect

   (3) Statement I correct and statement II incorrect

   (4) Statement I incorrect and statement II correct

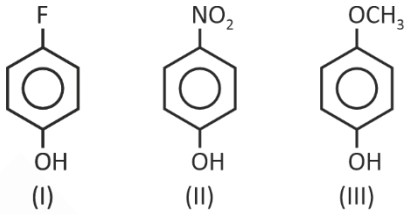
Answer (1)

Sol. Order of ligand strength $Cl^- > Br^-$

$[CoCl_4]^{2-}$ absorb more energy than $[CoBr_4]^{2-}$

$[CoCl_4]^{2-}$ has more CFSE value due to strength of ligand.

3. Arrange in increasing order of acid strength.



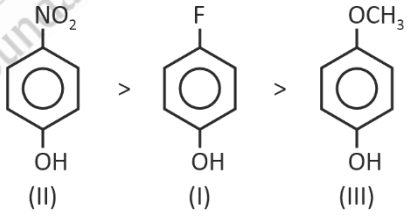
   (1) (II) > (I) = (III)

   (2) (I) > (II) > (III)

   (3) (II) > (I) > (III)

   (4) (II) = (I) > (III)

Answer (3)

Sol. K_a order 

4. Which of the following undergo nitration at fastest rate?

   (1) $C_6H_5NO_2$

   (2) $C_6H_5CH_3$

   (3) C_6H_5COOH

   (4) C_6H_5Br

Answer (2)

Sol. Methyl group is activating group

